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If you use ChatGPT or any other AI tool, your work will not be reviewed or accepted.

Data Processing Programming

Midterm Question

Version A	Questions
Park Youngjae	1. Explain the concept of "Data Cleaning" and its critical role in data preprocessing. Provide examples of common data cleaning operations. 2. Describe the various methods of handling missing data in datasets and the implications of each method. 3. Write a Python function using NumPy to calculate the mean and standard deviation of a given array. Your function should return both values. 4. Using Pandas, write a script to load a CSV file and filter out rows where any cell has a NaN value. 5. A DataFrame df has a column City with inconsistent capitalization: "Seoul", "seoul", "SEOUL", "Busan", "busan". Tasks: Normalize the City names so that all names start with an uppercase letter followed by lowercase letters. Print the cleaned column.
ABDUJALILOV	
JAMSHID	
ZHOU ZIXUAN	
K ZIN PO HTET	
KHOLBOEV ASILBEK	
BATSAIKHAN AZZAYA	
LOPCHAN NISHANT	
LAW KERYN SOK LUN	

Version B	Questions
AMIROV ASLIDDIN	1. Explain the term "Data Normalization" and its importance in the context of preparing data for machine learning models. 2. Discuss the importance of data types in NumPy and how they can affect computation in scientific computing. 3. Write a Python function using NumPy to calculate the median and interquartile range of a given array. Your function should return both values. 4. Using Pandas, write a script to load a CSV file and replace all NaN values with the mean of their respective columns. 5. Demonstrate how to create a 3D NumPy array from nested lists and print its dimensions.
XIAO JIAN	
BAATARCHULUUN	
UYANGA	
CAI YUPENG	
BEKMIRZAEV	
SALOHIDDIN	
SULTONOV BEHRUZ	
PARAJULI VISION	
LIU YANZHAO	

Version C	Questions
YANG XUEHAO	1. Explain the differences between the data structures Series and DataFrame in Pandas. 2. Explain what a NumPy array is and how it differs from a Python list in terms of performance and functionality. 3. Write a Python function using NumPy to perform a matrix multiplication between two given 2D arrays and return the result. 4. Using Pandas, write a script to load a JSON file into a DataFrame and print the first five rows. 5. You are given a DataFrame df with columns: Name, Age, Salary. Remove all rows where both Age and Salary are missing. Fill missing Age values with the average age. Fill missing Salary values with 0. Print the cleaned DataFrame.
OTAMIRZAEV ABDULLA	
DASHTSENGEL TELMEN	
CUI MENGYU	
AKHMADALIEV RUZIBOY	
POUDEL SHARMA BIPIN	
CHHETRI SABHYATA	
LIM XIN YU	

Version D	Questions
ZHOU SIYU	1. What is "Data Transformation"? Discuss its role in data preprocessing with examples of common transformation techniques. 2. What is Pandas in Python? Discuss its significance in data manipulation and analysis. 3. Write a Python function using NumPy to calculate the maximum and minimum values of a given array. Your function should return both values. 4. Using Pandas, write a script to load a CSV file and compute the sum of each column. 5. Given a CSV file named sales.csv with columns: Product, Price, and Quantity, write a Pandas program to: Load the CSV file into a DataFrame. Calculate a new column called Total = Price *Quantity. Print the top 5 rows of the Data Frame.
KOMILJONOV RAHIMJON	
AKHMEDOV YUSUFJON	
YE SWAM YEE	
MAI TRONG KHANG	
CHAULAGAIN SUDIP	
BATNASAN ERDENESAMBUU	
PANDAY PRASHANT TAN ELDRIC	

Version E	Questions
CUI XIAOSEN	- What are the main steps in the data processing cycle? Briefly explain each step. - Explain the importance of data cleaning in the data processing pipeline. Provide examples of at least three common data cleaning tasks. - Demonstrate how to reshape a 1D NumPy array into a 2D array with 3 rows. - Create a 5×5 NumPy array of random numbers, convert it into a Pandas DataFrame, and then: Rename the columns as A, B, C, D, and E. Find the column with the highest average value and print its name. - Given a CSV file named employees.csv containing the columns: Name, Department, Salary, and JoiningDate, write a Pandas program to: Load the file into a DataFrame and convert JoiningDate to datetime. Remove rows with missing Salary values. Group the data by Department and calculate the average salary for each department. Print the department with the highest average salary.
AMARJARGAL OYUNZUL	
LIU HONGJUN	
SHUKIRILLAEV MUKHAMMADSODIK	
ACHHAMI SHITAL	
TURAEV LAZIZBEK	
YONG EVANS CHENG YU	

Version F	Questions
CHANG ALVIN CHUO WEN	- What are common strategies for handling missing data in a dataset? Explain the advantages and disadvantages of at least two methods. - Explain the difference between data cleaning and data transformation with examples. - Demonstrate how to create a 2D NumPy array from a list of lists and print its shape. - Drop all nan values from a 1D numpy array Input: np.array([1,2,3,np.nan,5,6,7,np.nan]) - You have a DataFrame df containing EmployeeID, Name, Department. Tasks: Identify and remove all duplicate rows based on EmployeeID. Print the total number of rows before and after removing duplicates.
CHANG JIAN RONG	
RAI PRASHNA	
ARYAL DIPIKA	
IBROKHIMJONOV FAYZULLO	
LAI YUNDA	
MUNGUNBAATAR CHINGUN	

Version G	Questions
WONG JUN LIN	- What are common strategies for handling missing data in a dataset? Explain the advantages and disadvantages of at least two methods. - Explain how you can filter and select specific rows or columns in a Pandas DataFrame. Provide examples of at least two different methods. - You have a DataFrame df with column Price stored as strings: ['1000', '2500', '3000', '4000']. Tasks: - Convert Price to numeric type (int). - Create a new column DiscountedPrice which is 90% of Price. - Print the DataFrame with the new column. - Demonstrate how to use slicing to extract a subarray from a 2D NumPy array. - Given a CSV file named employees.csv containing the columns: Name, Department, Salary, and JoiningDate, write a Pandas program to: - Load the file into a DataFrame and convert JoiningDate to datetime. - Remove rows with missing Salary values. - Group the data by Department and calculate the average salary for each department. - Print the department with the highest average salary.
OCHILOV AKHMADJON	
ABDUKHOLIKOV	
MUHAMMADSODIK	
KHANAL PRAKASH	
TANG TIAN	
HOANG VAN HAI	